Computer Networks

Lecture- 07

Recap of Lecture 05 & 06

- The OSI Model
- Layered task
- Peer-to-Peer communication
- OSI Layers

- TCP/IP Reference model

Overview

- Transmission Media and its Types
- Guided Media
  - Twisted Pair
    - UTP
    - STP
  - Coaxial Cable
  - Optical Fiber
    - Multimode
    - Single mode

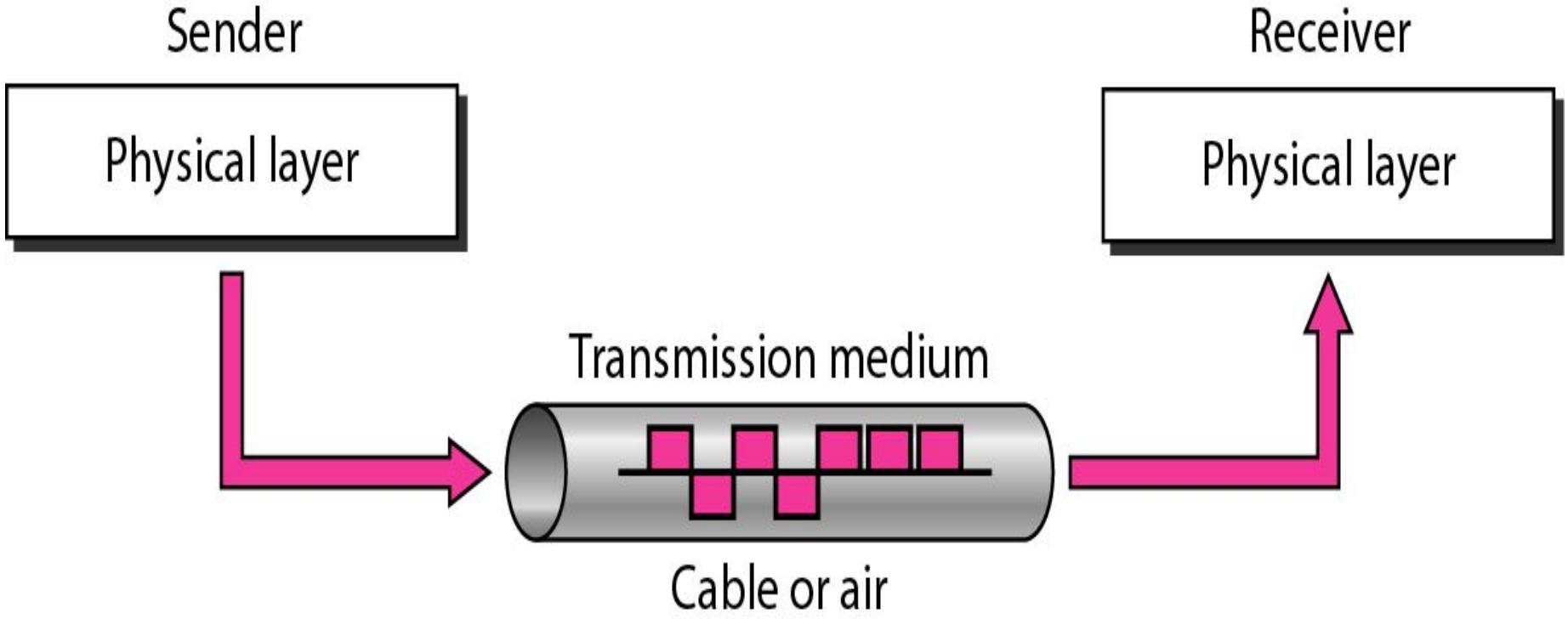
Transmission Media

- Transmission media are actually located below the physical layer (Layer 0) and are directly controlled by the physical layer.

- A **transmission medium** can be broadly defined as anything that can **carry information** from a source to a destination.

- The air can also be used to convey the message in a smoke signal or semaphore.
- For a written message, the transmission medium might be a mail carrier, a truck, or an airplane.
- In data communications transmission medium is usually free space (air), metallic cable, or fiber-optic cable.

Transmission medium and physical layer

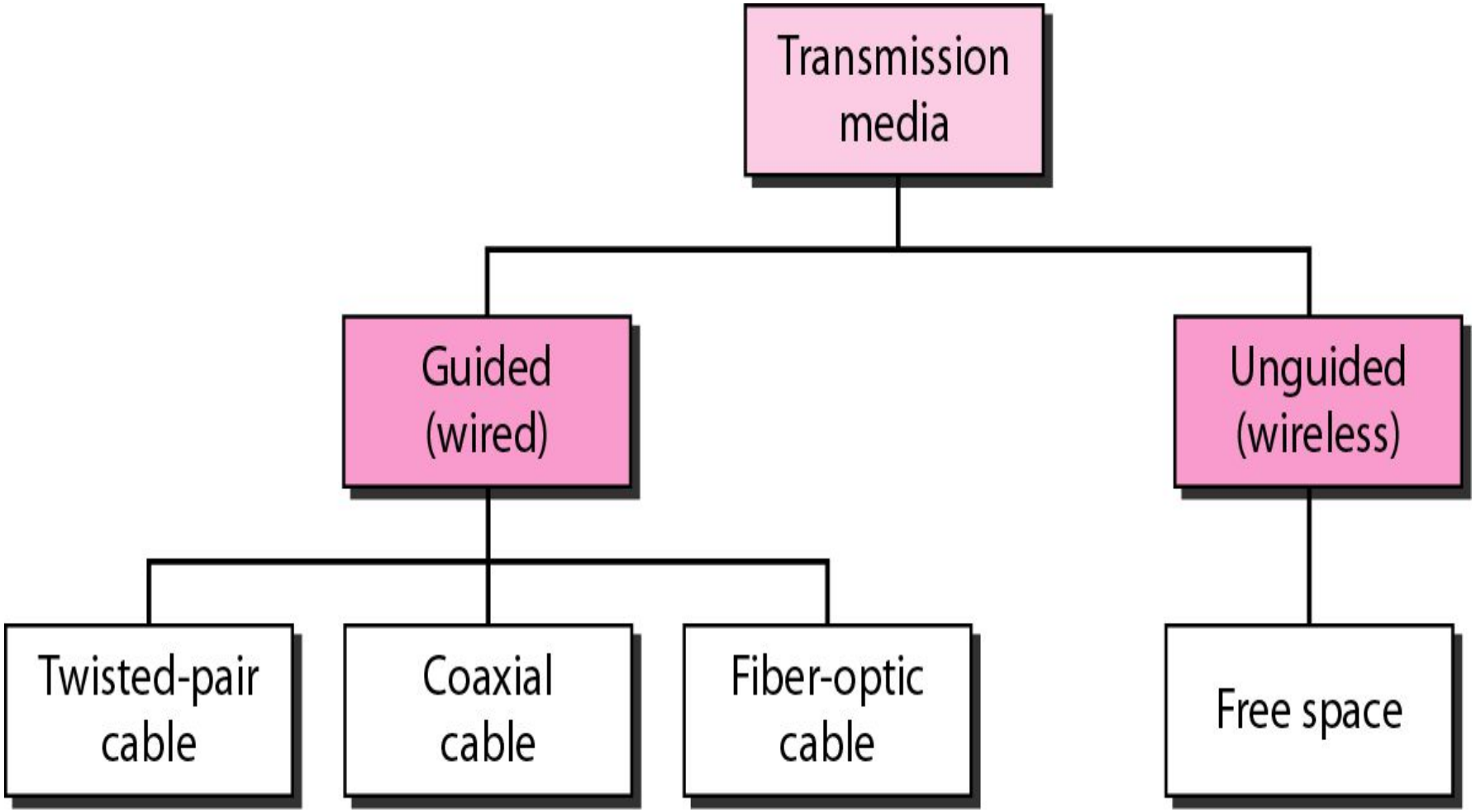


Transmission Media

- computers and other telecommunication devices use signals to represent data.
- These signals are transmitted from one device to another in the form of **electromagnetic energy**, which is propagated through transmission media.

- Electromagnetic energy, a combination of electric and magnetic fields vibrating in relation to each other, includes power, radio waves, infrared light, visible light, ultraviolet light, and X, gamma, and cosmic rays.

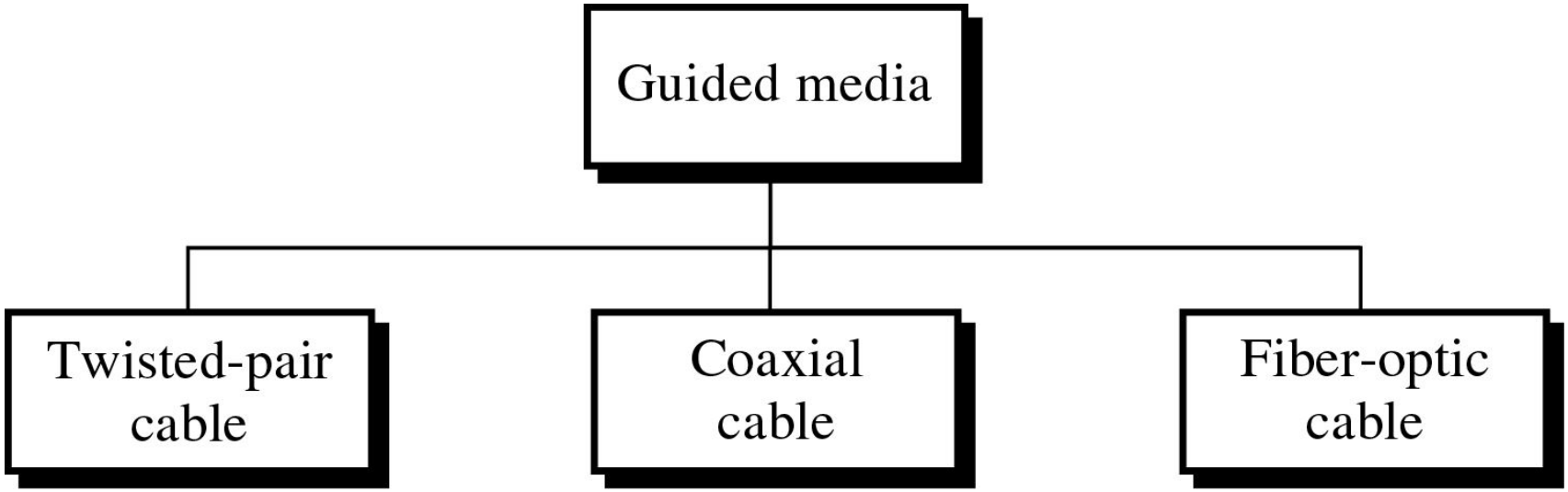
Classes of Transmission Media



Guided Media

- Provide a conduit from one device to another.
- A signal travelling along any of these media is directed and contained by the physical limits of the medium.

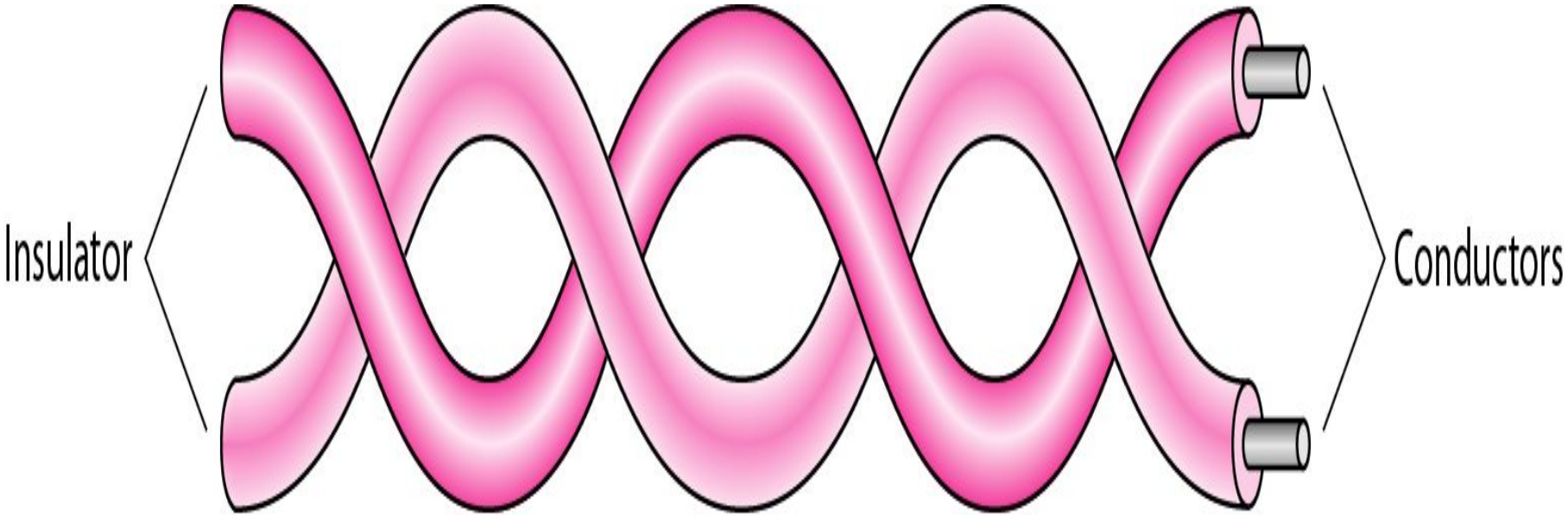
Categories of Guided Media



Twisted-Pair Cable

- A twisted pair consists of two conductors (normally copper), each with its own plastic insulation, twisted together.
- One of the wires is used to carry signals to the receiver, and the other is used only as a ground reference.
- The receiver uses the difference between the two.
- In addition to the signal sent by the sender on one of the wires, interference (noise) and crosstalk may affect both wires and create unwanted signals.

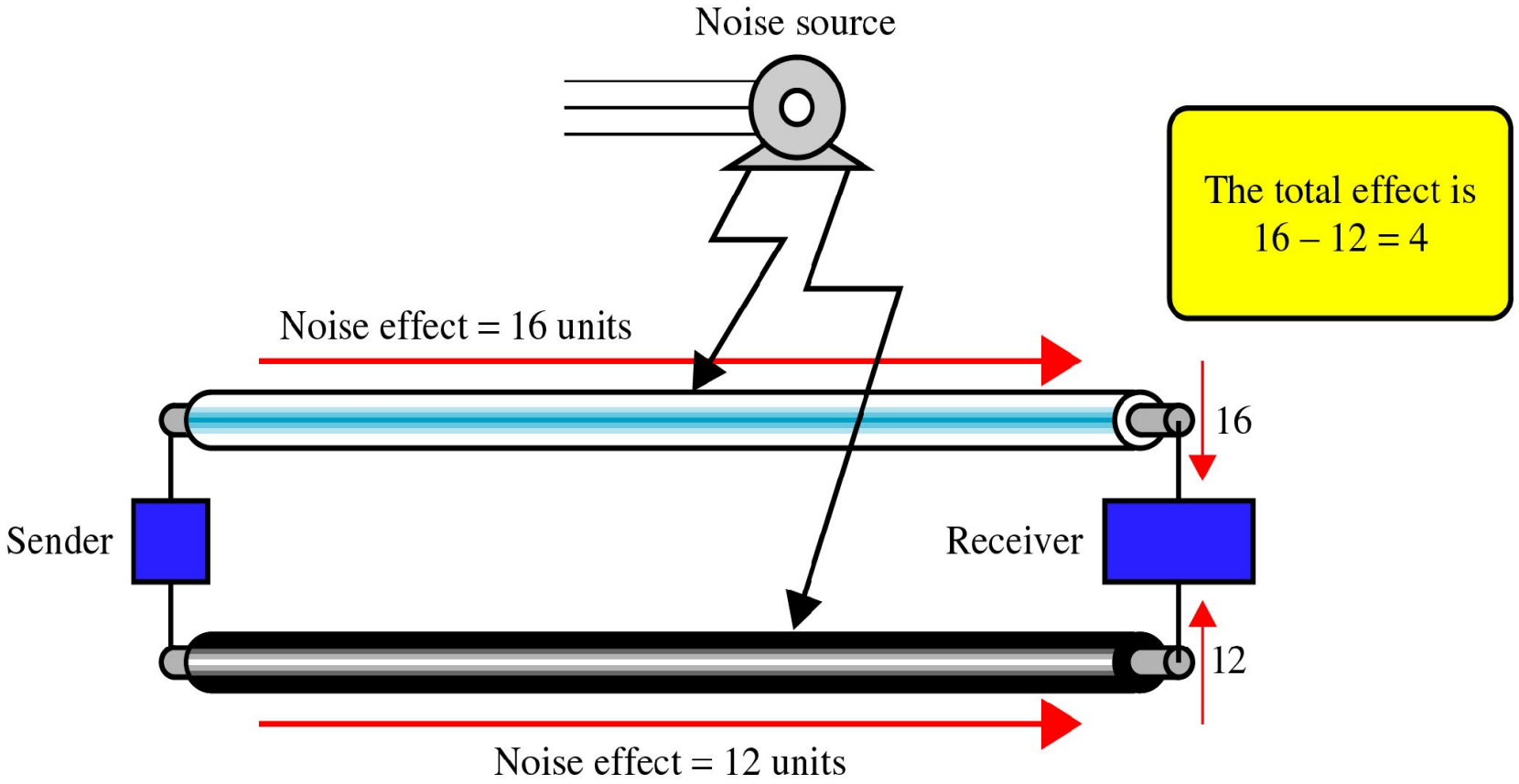
Twisted-Pair Cable



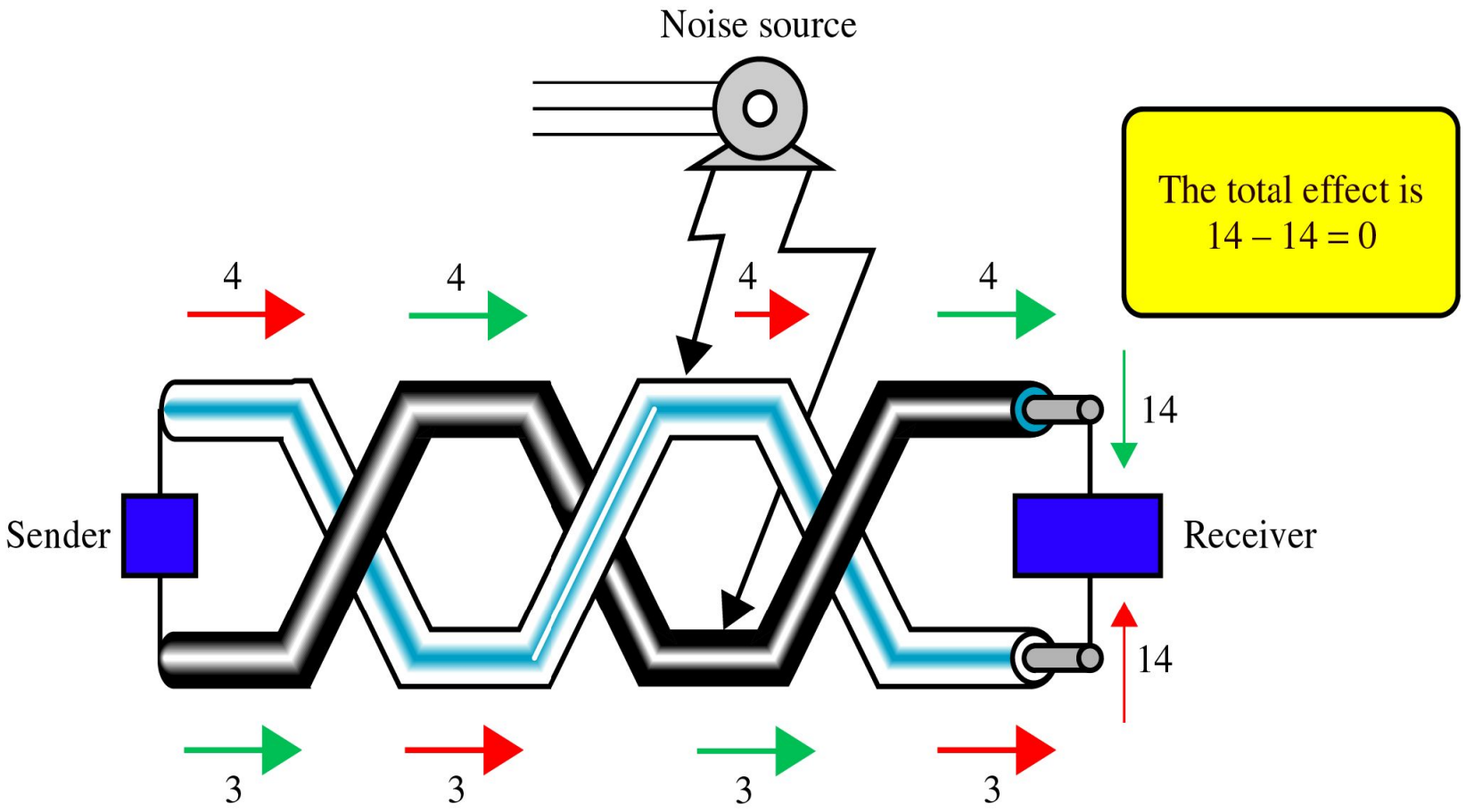
Twisted-Pair Cable

- If the two wires are parallel, the effect of these unwanted signals is not the same in both wires because they are at different locations relative to the noise or crosstalk sources (e,g., one is closer and the other is farther).This results in a difference at the receiver.
- By twisting the pairs, a balance is maintained.
- Twisting makes it probable that both wires are equally affected by external influences (noise or crosstalk).
- This means that the receiver, which calculates the difference between the two, receives no unwanted signals.
- The unwanted signals are mostly canceled out.
- The number of twists per unit of length (e.g., inch) has some effect on the quality of the cable.

Parallel Flat Wire



Noise Effect on Twisted-Pair

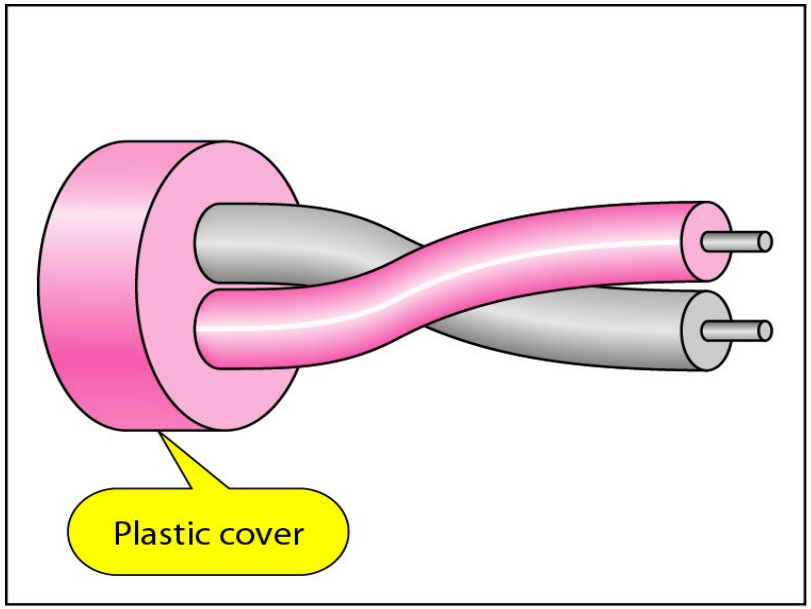


Twisted-Pair Cable

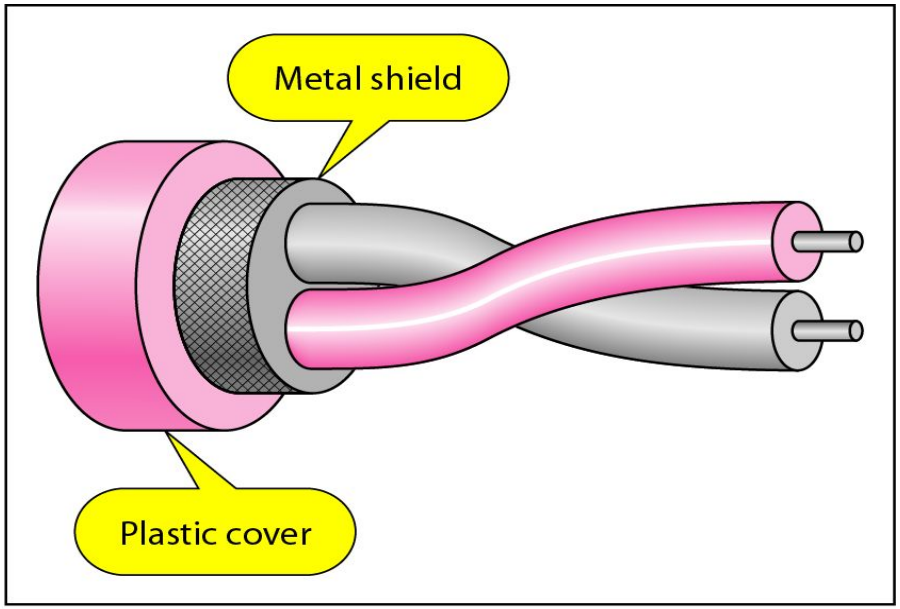
- Two forms
  1. Unshielded twisted-pair (UTP)
  2. Shielded twisted-pair (STP)

UTP Cable

- One or more pairs of insulated wires within an enclosing insulation sheath.
- Only one outer protection sheath.
- Can be one, two, four, six, and eight pair wiring (same for STP).

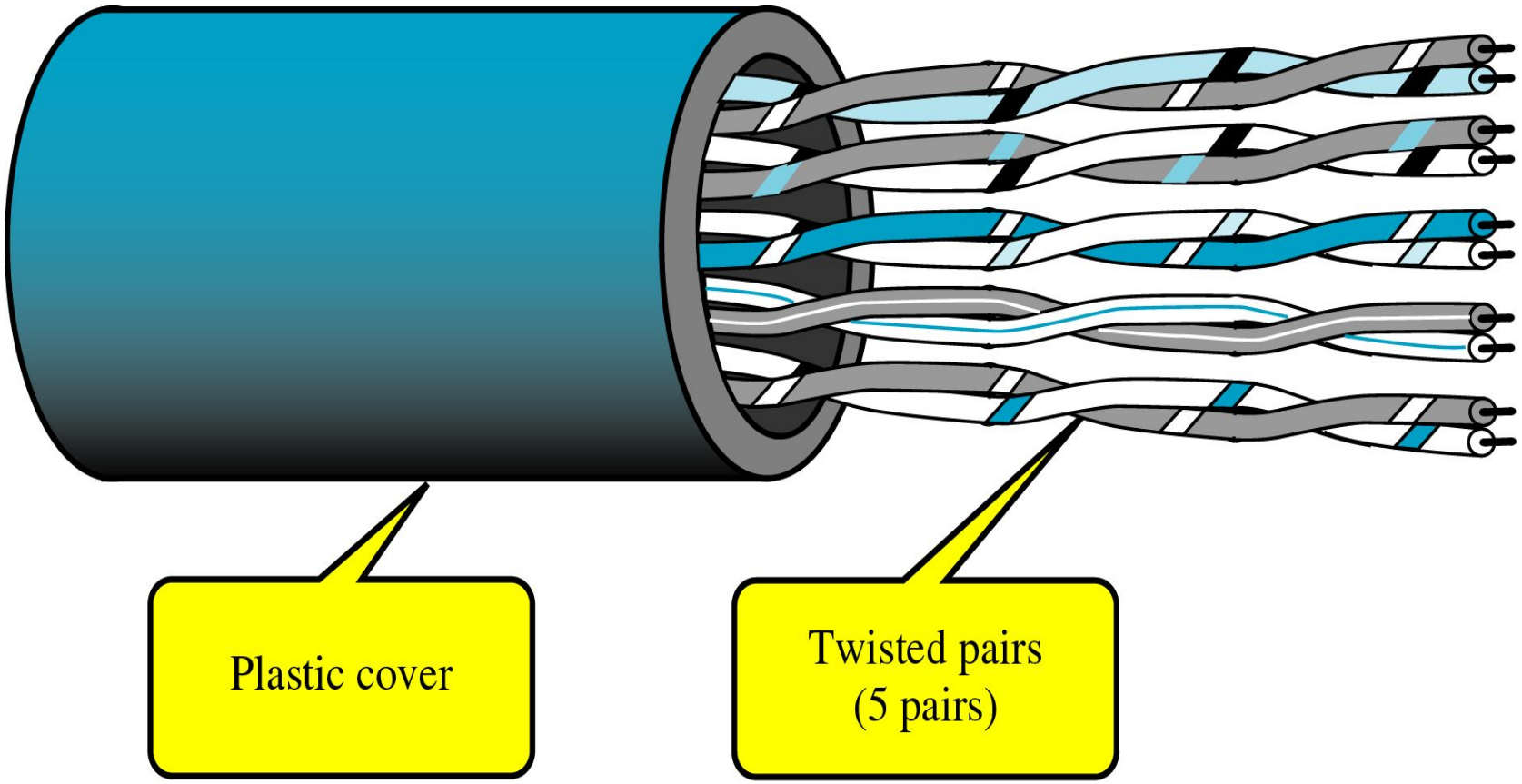


a. UTP



b. STP

UTP Cable

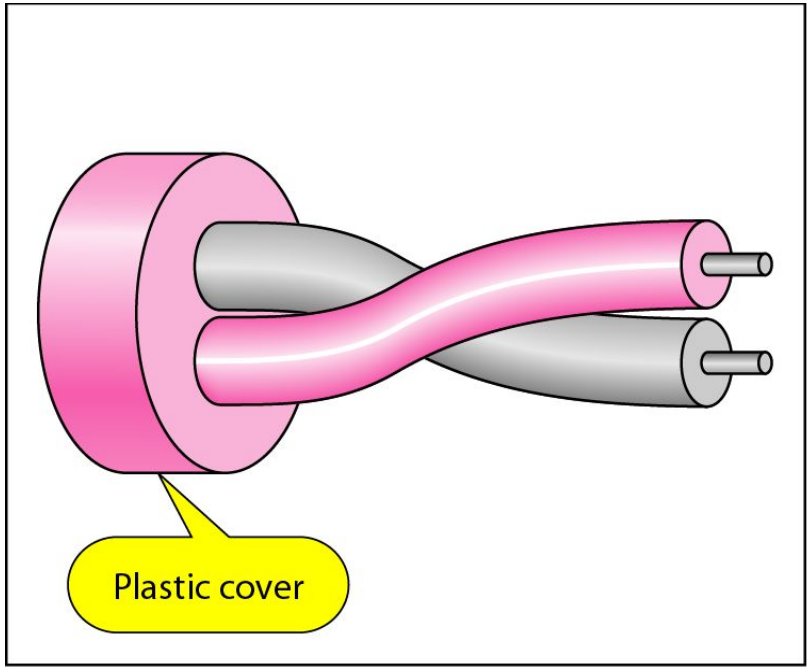


Advantage of UTP

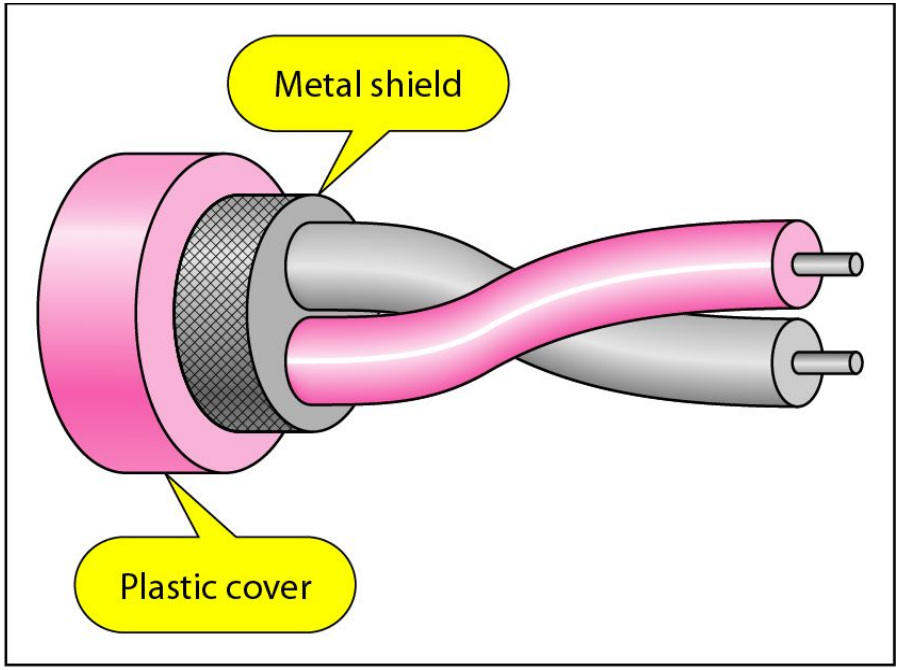
- Advantages of UTP are:
  - Cost
  - Ease of Use

Shielded Twisted Pair (STP) Cable

- UTP cable + metal casing (foil or braided-mesh)
- The metal casing prevents EM noise

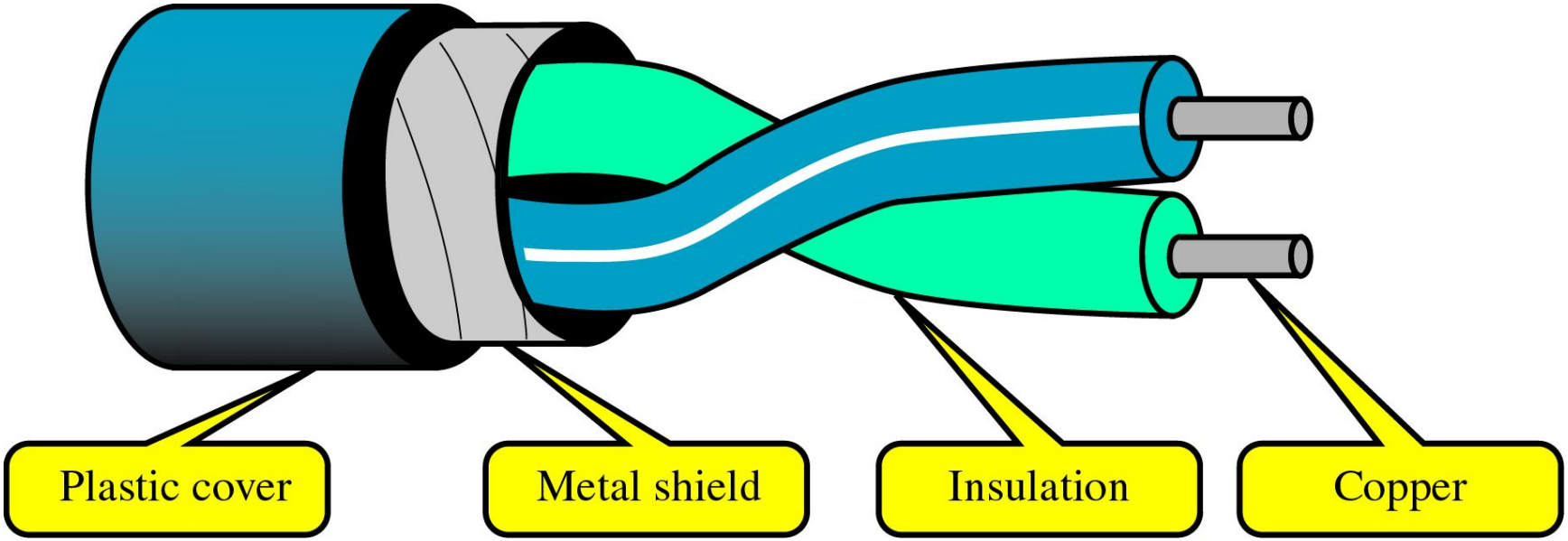


a. UTP



b. STP

STP Cable

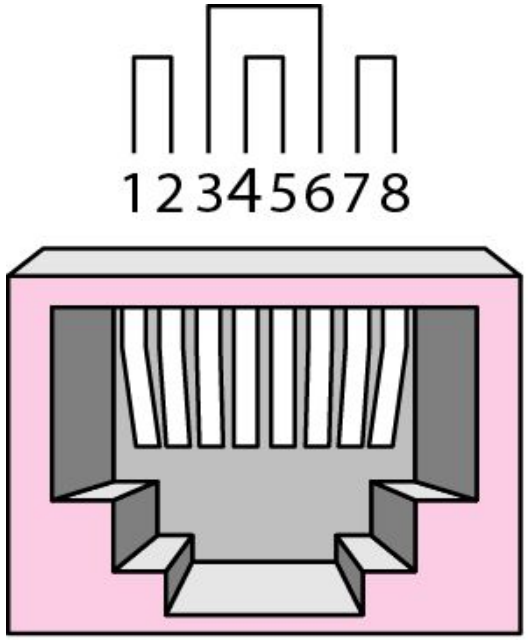


Categories of UTP Cable

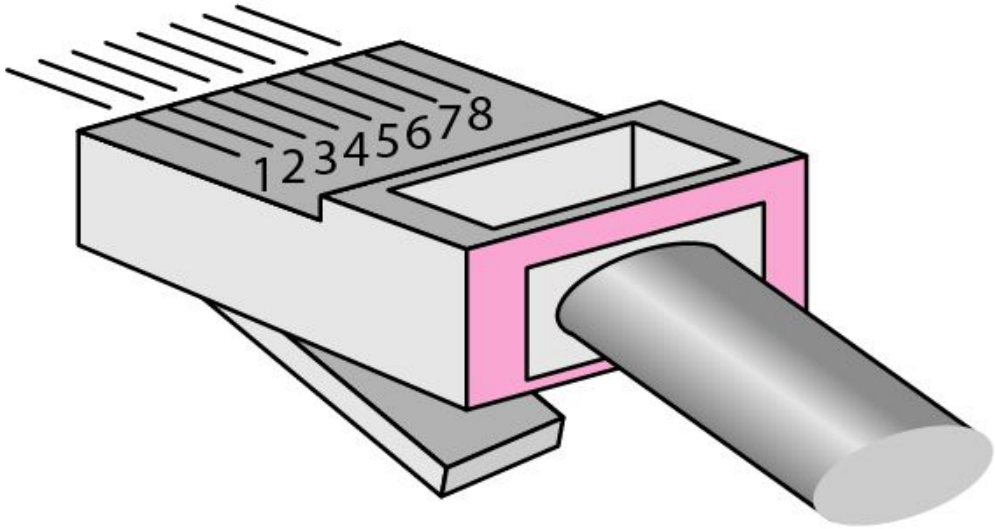
Category	*Specification*	*Data Rate (Mbps)*	*Use*
1	Unshielded twisted-pair used in telephone	< 0.1	Telephone
2	Unshielded twisted-pair originally used in T-lines	2	T-1 lines
3	Improved CAT 2 used in LANs	10	LANs
4	Improved CAT 3 used in Token Ring networks	20	LANs
5	Cable wire is normally 24 AWG with a jacket and outside sheath	100	LANs
5E	An extension to category 5 that includes extra features to minimize the crosstalk and electromagnetic interference	125	LANs
6	A new category with matched components coming from the same manufacturer. The cable must be tested at a 200-Mbps data rate.	200	LANs
7	Sometimes called SSTP (shielded screen twisted-pair). Each pair is individually wrapped in a helical metallic foil followed by a metallic foil shield in addition to the outside sheath. The shield decreases the effect of crosstalk and increases the data rate.	600	LANs

UTP Connectors

- UTP uses RJ-45 Connector for computer networks (RJ-Registered Jack)

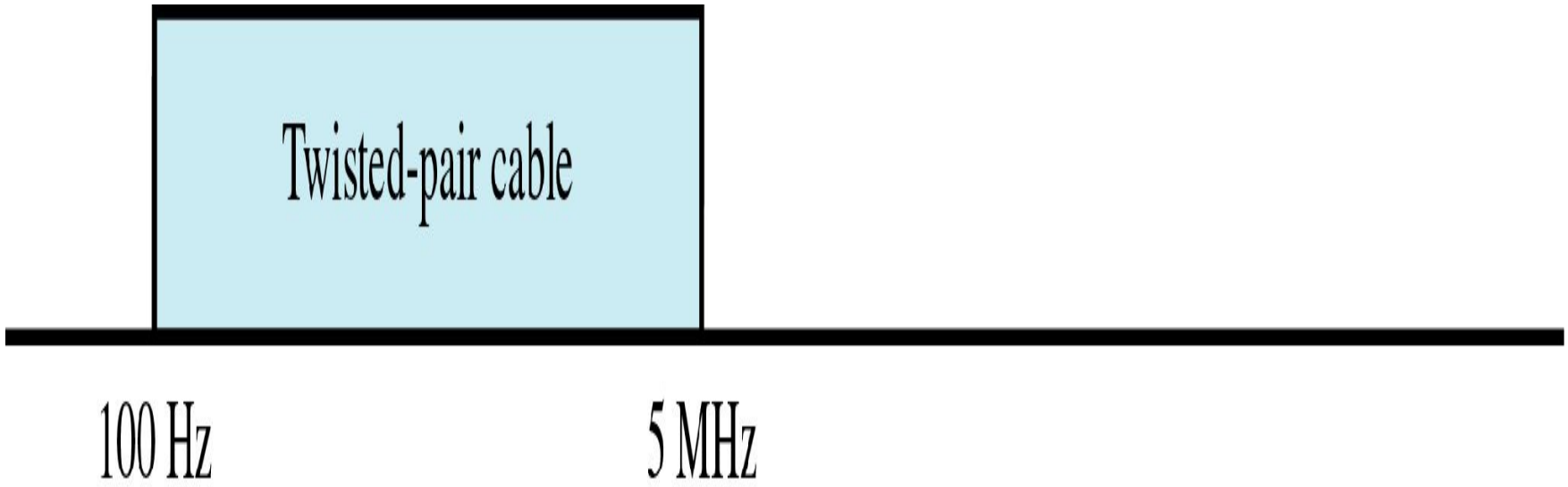


RJ-45 Female



RJ-45 Male

Frequency of Twisted-Pair Cable

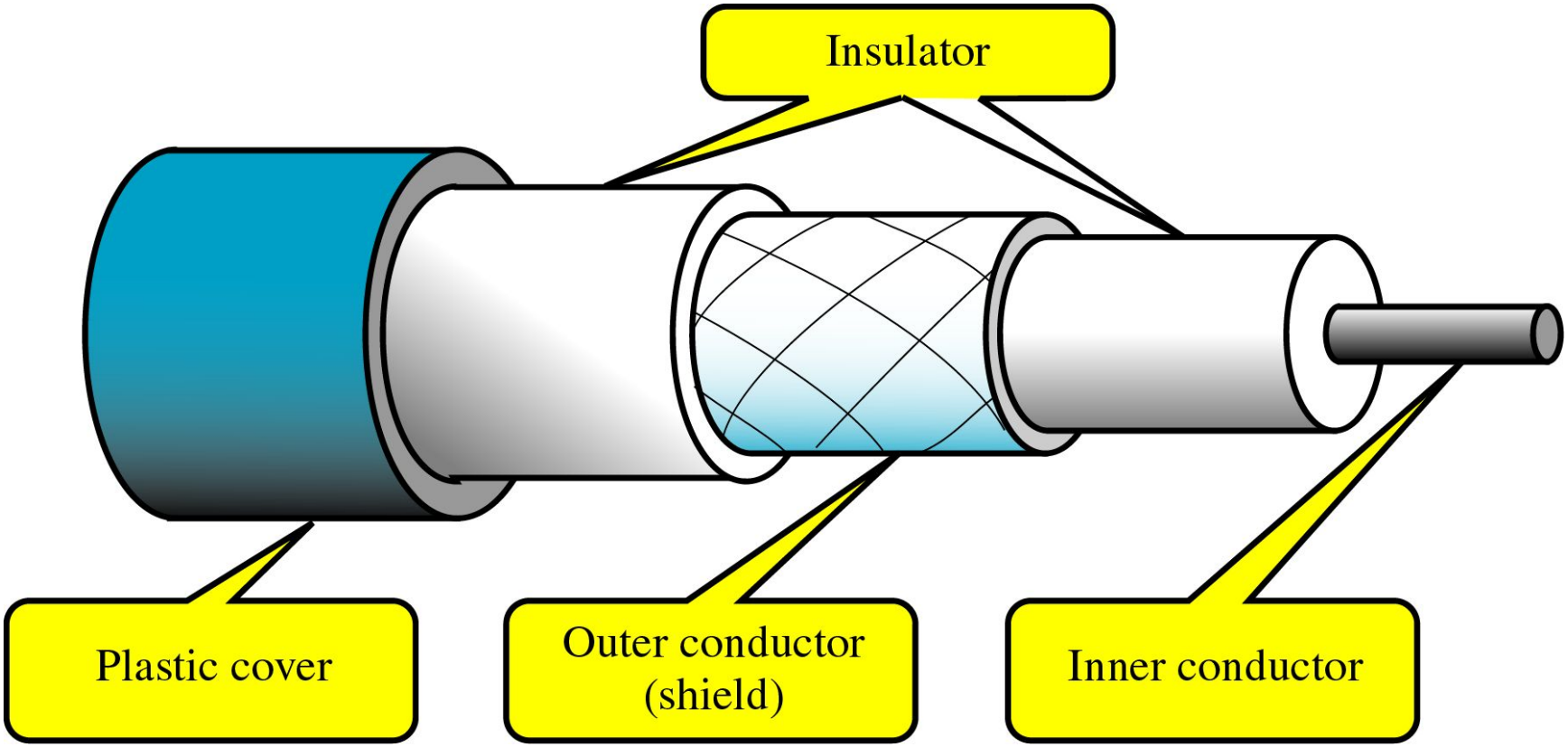


Coaxial Cable

Coaxial Cable

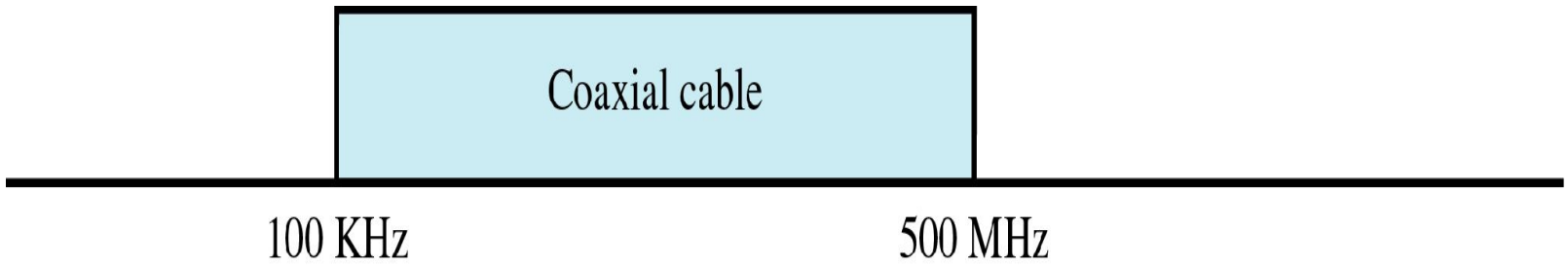
- Coaxial cable (or coax) carries signals of higher frequency ranges than those in twisted pair cable.
- Instead of having two wires, coax has a central core conductor of solid or stranded wire (usually copper) enclosed in an insulating sheath, which is, in turn, encased in an outer conductor of metal foil, braid, or a combination of the two.
- The outer metallic wrapping serves both as a shield against noise and as the second conductor, which completes the circuit.
- This outer conductor is also enclosed in an insulating sheath, and the whole cable is protected by a plastic cover.

Coaxial Cable



Frequency Range of Coaxial Cable

- Coaxial cable (coax) carries signals of higher frequency ranges than twisted pair cable.



Coaxial Cable Standards

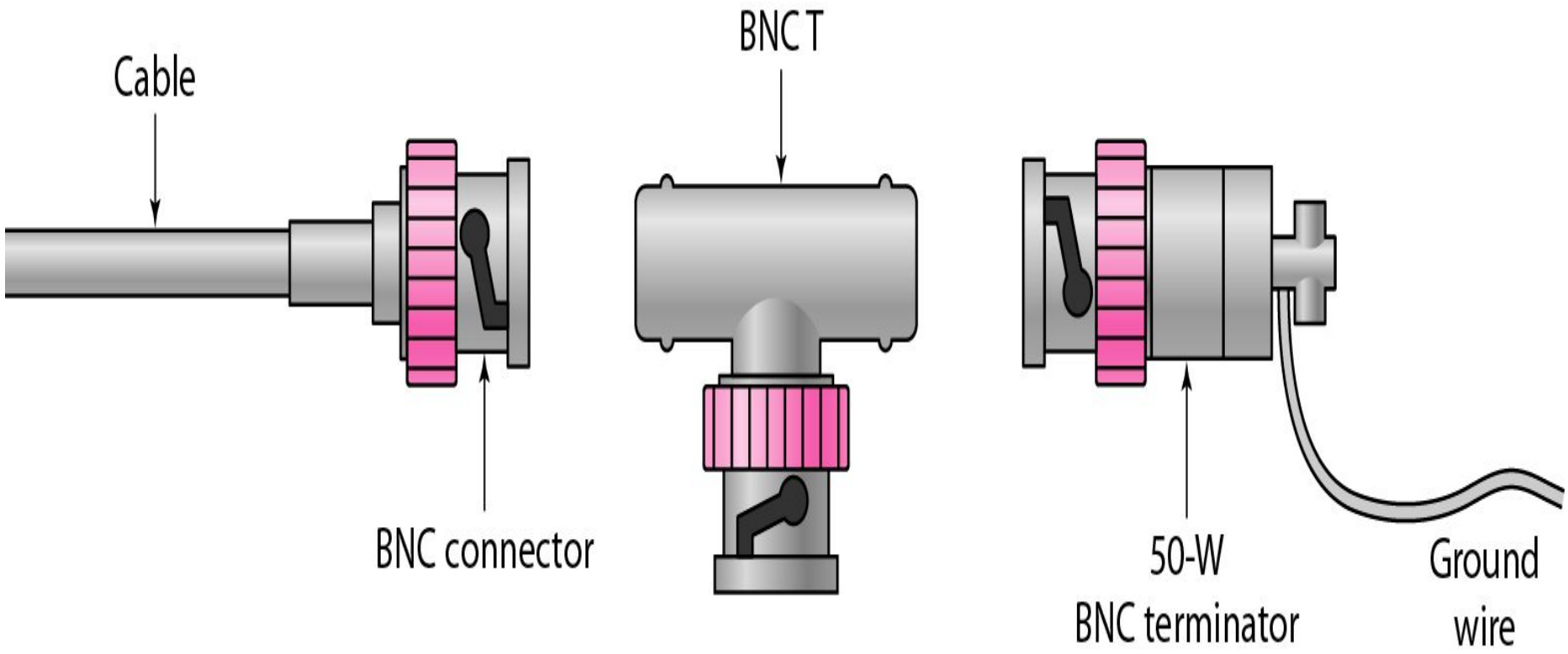
- Coaxial cables are categorized by their **radio government** (RG) ratings.
- Each RG number denotes a unique set of physical specifications, including the wire gauge of the inner conductor, the thickness and type of the inner insulator, the construction of the shield, and the size and type of the outer casing.
- Each cable defined by an RG rating is adapted for a specialized function, as shown in Table.

Category	*Impedance*	*Use*
RG-59	75 Ω	Cable TV
RG-58	50 Ω	Thin Ethernet
RG-11	50 Ω	Thick Ethernet

Coaxial Cable Connectors

- The most common type of connector used today is the **Bayone-Neill-Concelman (BNC), connector**.
- The **BNC connector** is used to connect the end of the cable to a device, such as a TV set.
- The **BNC T connector** is used in Ethernet networks to branch out to a connection to a computer or other device.
- The **BNC terminator** is used at the end of the cable to prevent the reflection of the signal.

Coaxial Cable Connectors



Application of Coaxial Cable

1. Started in Analog telephone network (10,000 voice signals)
2. Later used in Digital Telephone network (600 Mbps)
3. Cable TV Network also uses coaxial cable
4. Traditional Ethernet LANs
   - 10Base2 (Thin net) uses RG-58 cable
   - 10Base5 (Thick Net) uses RG-11 cable

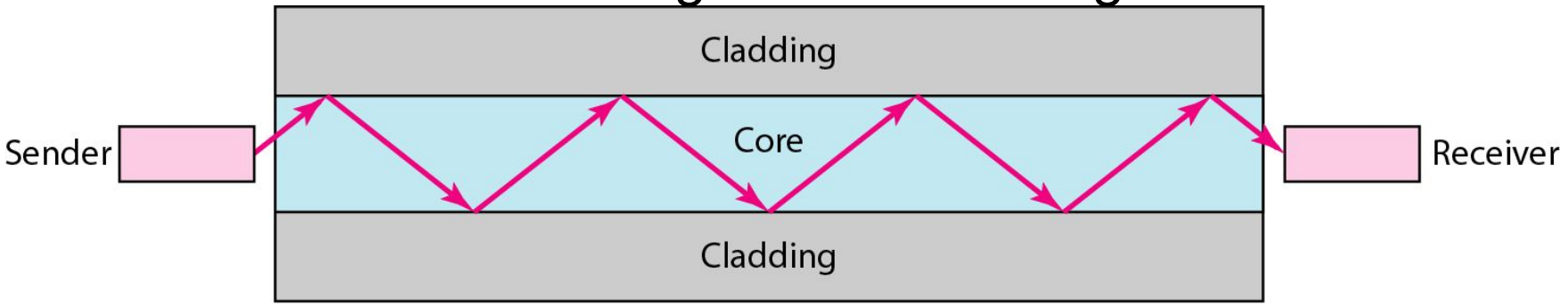
Optical Fiber

Optical Fiber

- A fiber-optic cable is made of glass or plastic and transmits signals in the form of light.
- Light travels in a straight line as long as it is moving through a single uniform substance.
- If a ray of light traveling through one substance suddenly enters another substance (of a different density), the ray changes direction.
- Figure shows how a ray of light changes direction when going from a more dense to a less dense substance.

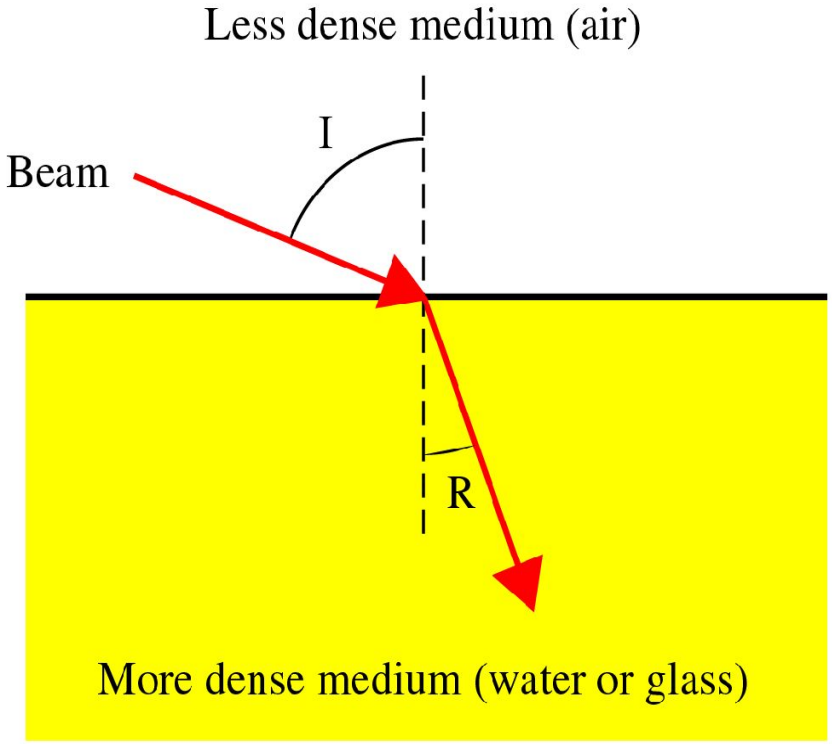
Optical Fiber

- Optical fibers use reflection to guide light through a channel.
- A glass or plastic core is surrounded by a cladding of less dense glass or plastic.
- The difference in density of the two materials must be such that a beam of light moving through the core is reflected off the cladding instead of being refracted into it.

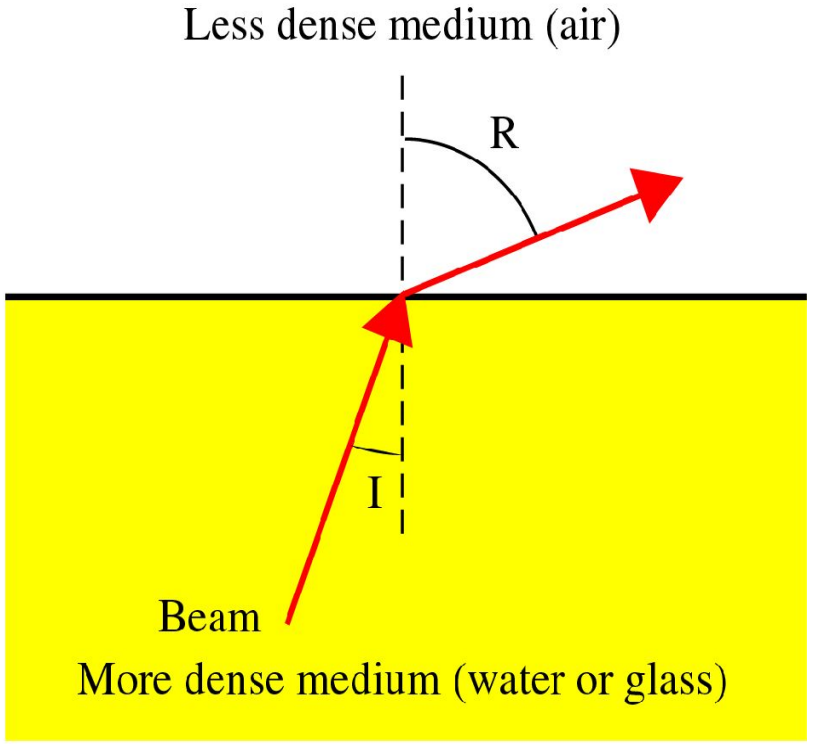


Refraction

- Change of direction of a light ray
  - When a ray of light enters another substance, its speed changes abruptly, causing the ray to change direction



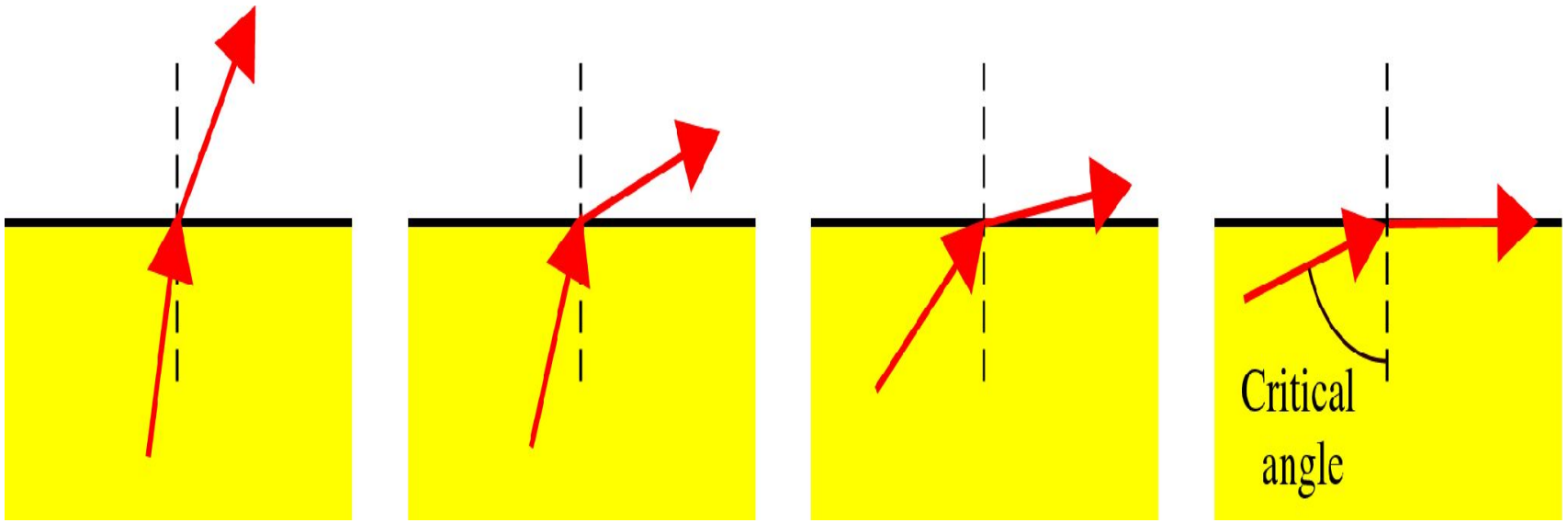
a. From less dense to more dense medium



b. From more dense to less dense medium

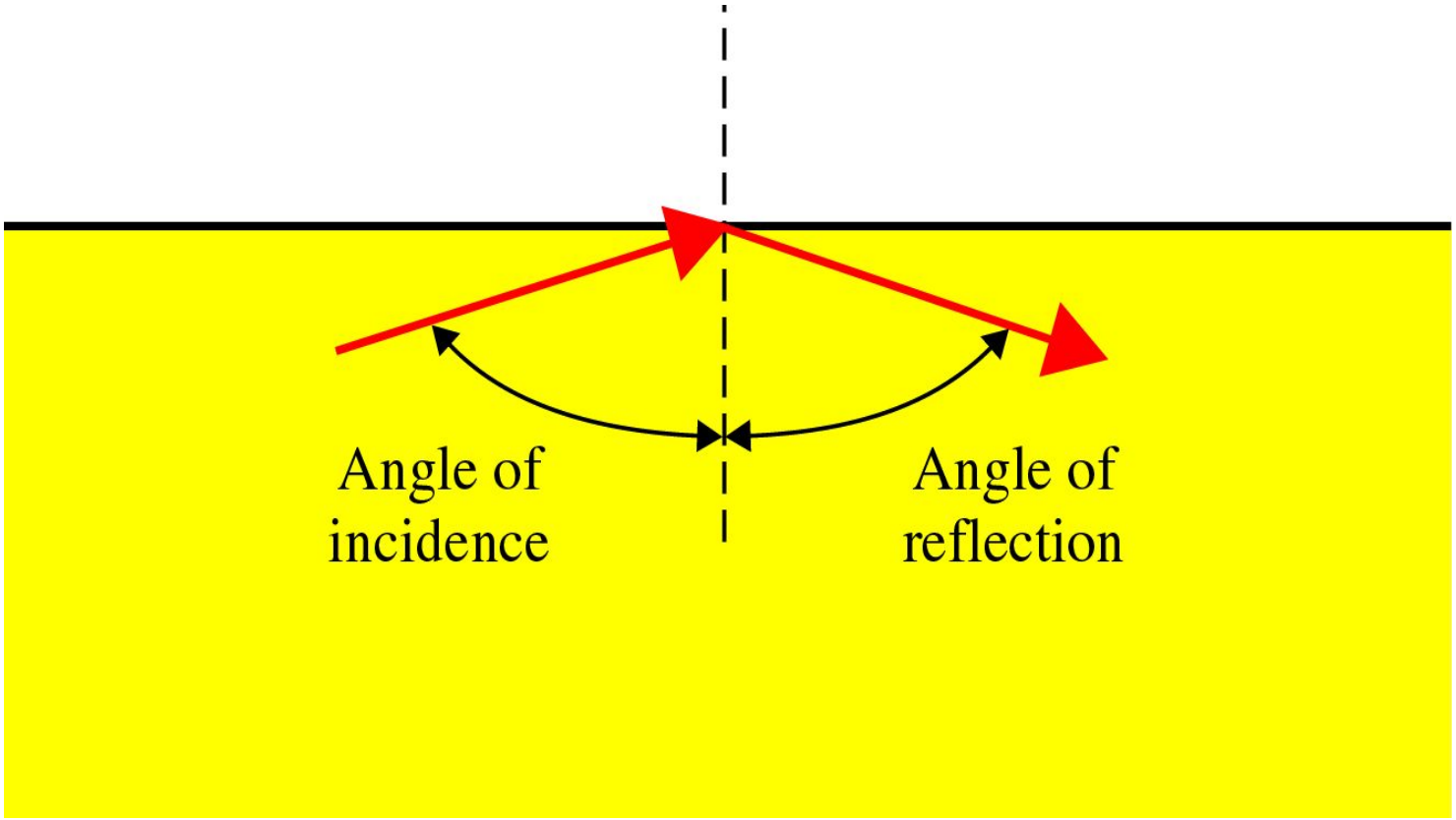
Critical Angle

- The incident angle at which the refracted angle is 90 degrees

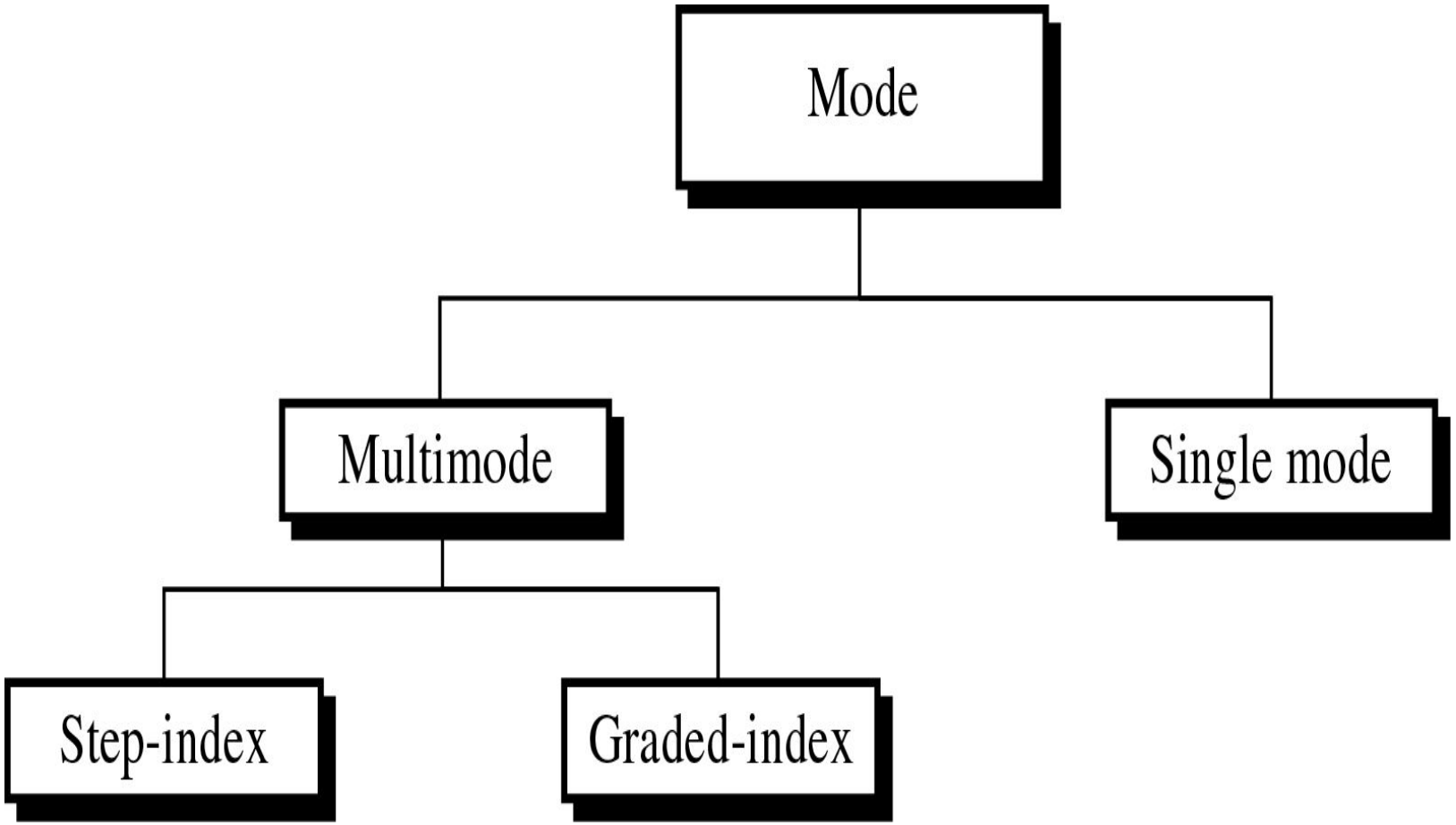


Reflection

- A phenomenon which occurs when the incident angle is greater than the critical angle
- Light no longer passes into the less dense medium
- Incident angle = reflected angle



Propagation Modes



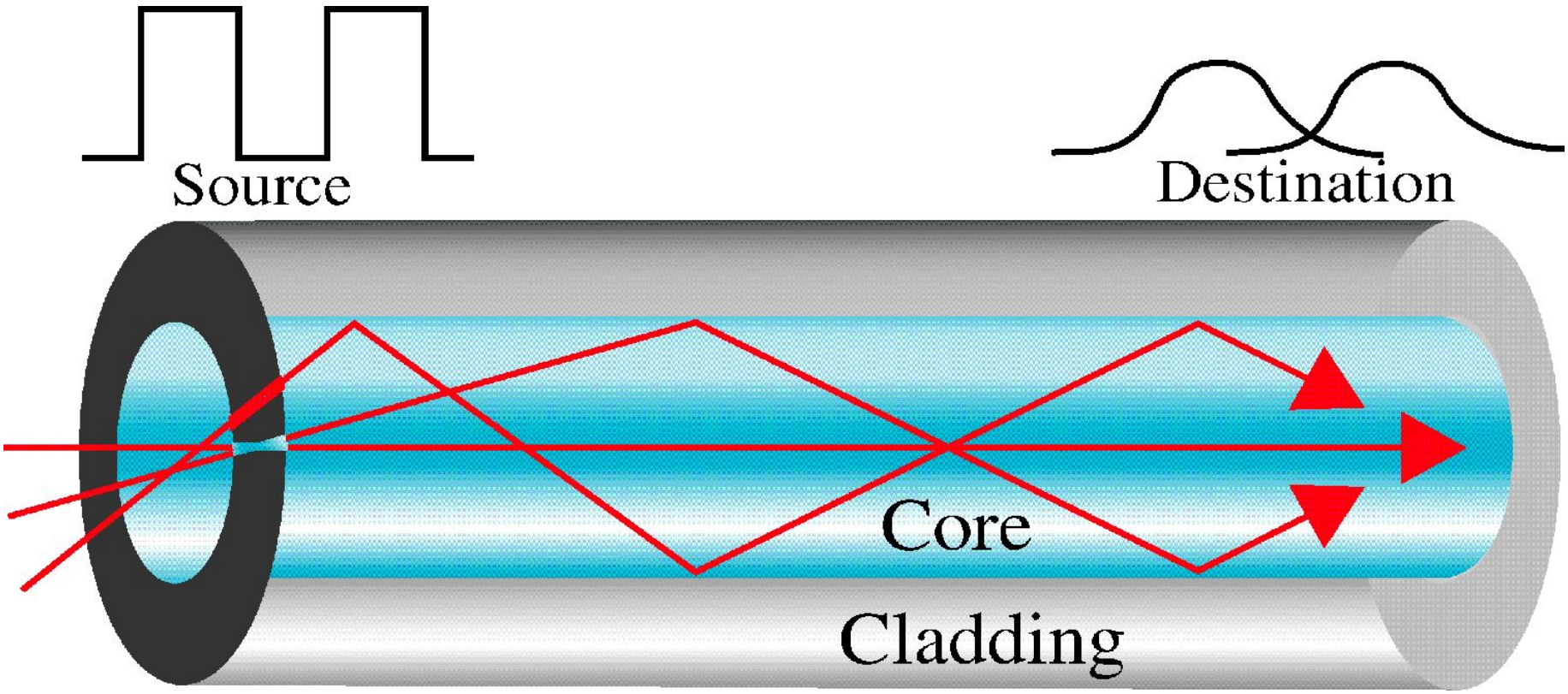
Multimode Fiber

- Multimode is so named because multiple beams from a light source move through the core in different paths.
- How these beams move within the cable depends on the structure of the core.

Multimode Step-Index Fiber

- In multimode step-index fiber, the density of the core remains constant from the center to the edges.
- A beam of light moves through this constant density in a straight line until it reaches the interface of the core and the cladding.
- At the interface, there is an abrupt change due to a lower density; this alters the angle of the beam's motion.
- The term step index refers to the suddenness of this change, which contributes to the distortion of the signal as it passes through the fiber.

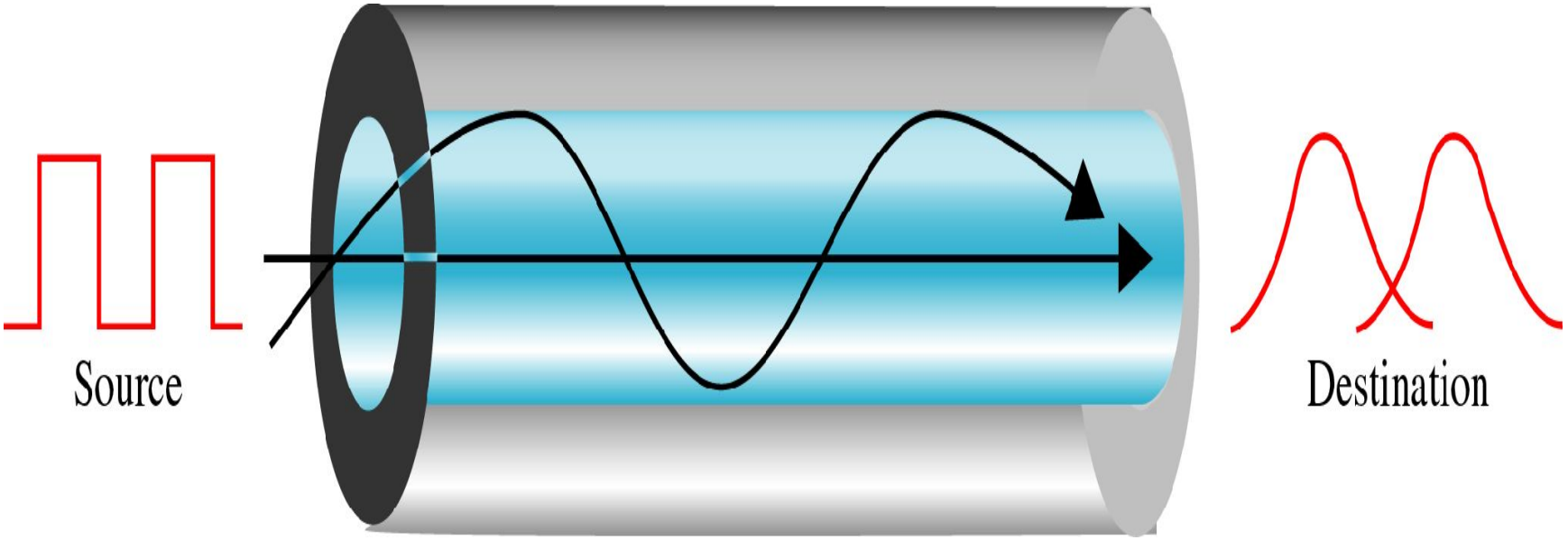
Multimode Step-Index Fiber



Multimode Graded-Index Fiber

- A second type of fiber, called multimode graded-index fiber, decreases this distortion of the signal through the cable.
- The word index here refers to the index of refraction.
- As we saw above, the index of refraction is related to density.
- A graded-index fiber, therefore, is one with varying densities.
- Density is highest at the center of the core and decreases gradually to its lowest at the edge.
- Figure shows the impact of this variable density on the propagation of light beams.

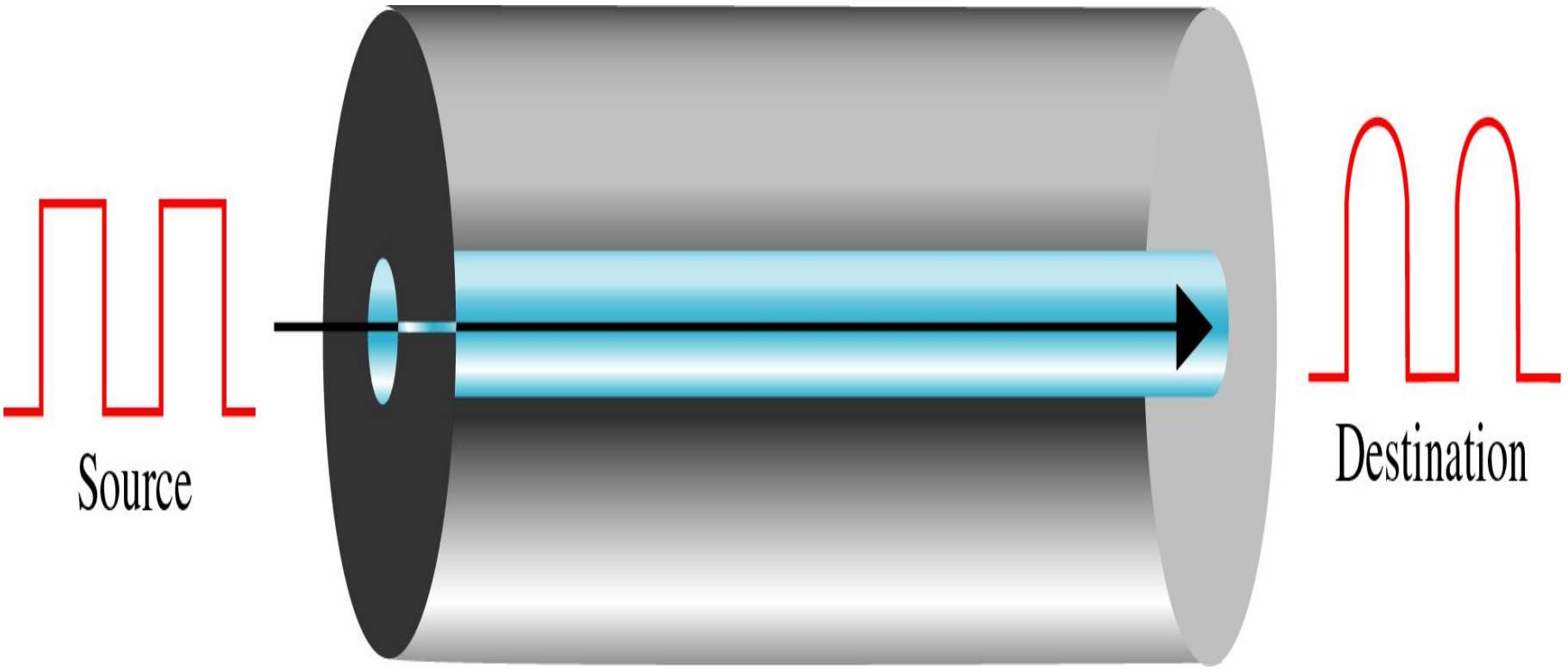
Multimode Graded-Index Fiber



Single Mode Fiber

- Single-mode uses step-index fiber and a highly focused source of light that limits beams to a small range of angles, all close to the horizontal.
- The single-mode fiber itself is manufactured with a much smaller diameter than that of multimode fiber, and with substantially lower density (index of refraction).
- The decrease in density results in a critical angle that is close enough to 90° to make the propagation of beams almost horizontal.
- In this case, propagation of different beams is almost identical, and delays are negligible.
- All the beams arrive at the destination "together" and can be recombined with little distortion to the signal.

Single Mode Fiber



Fiber Sizes

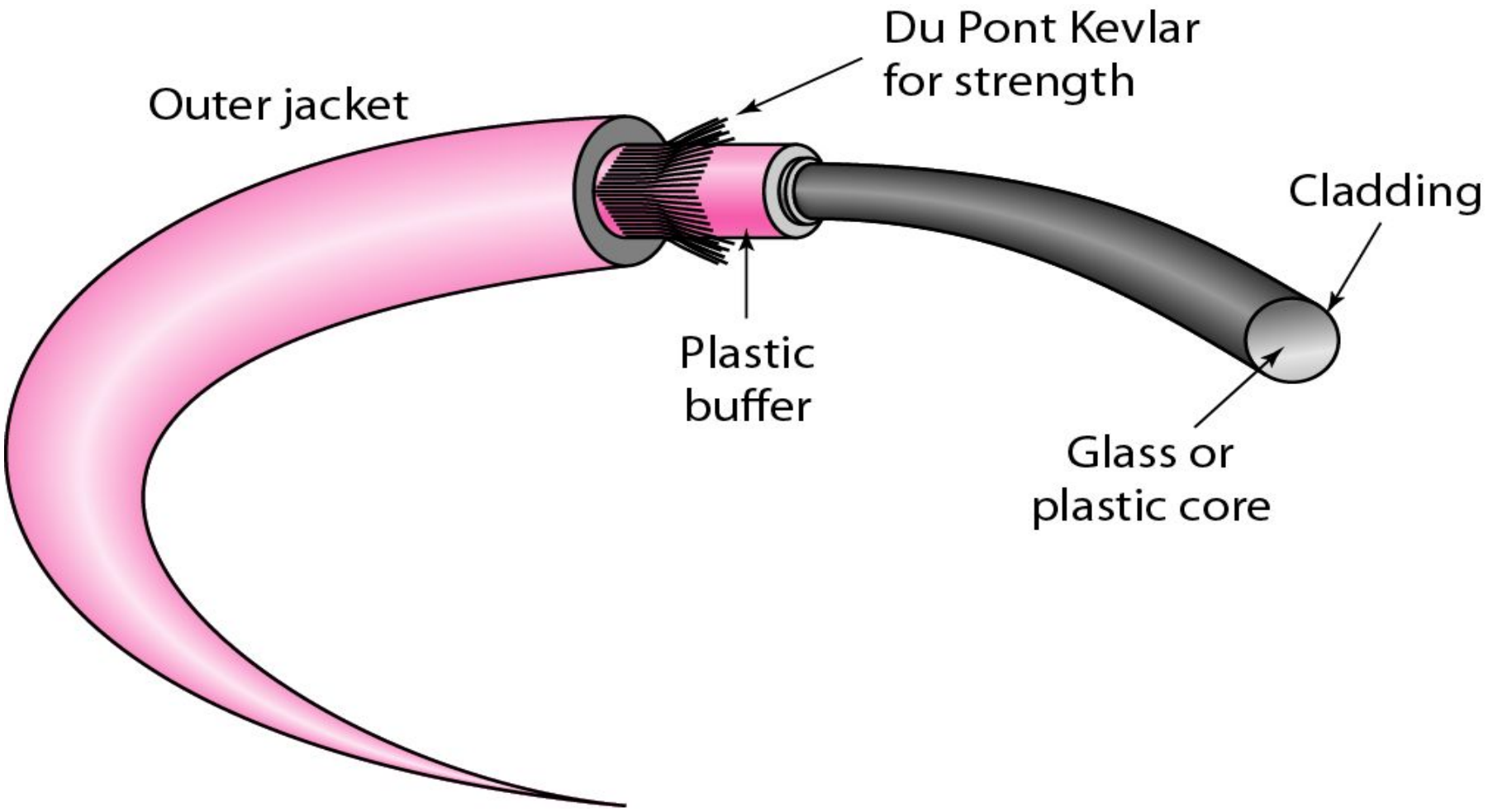
- Defined by the ratio of the core diameter to the cladding diameter, expressed in microns Optical fibers are defined by the ratio of the diameter of their core to the diameter of their cladding, both expressed in micrometers.

Type	*Core (µm)*	*Cladding (µm)*	*Mode*
50/125	50.0	125	Multimode, graded index
62.5/125	62.5	125	Multimode, graded index
100/125	100.0	125	Multimode, graded index
7/125	7.0	125	Single mode

Cable Composition

- Figure shows the composition of a typical fiber-optic cable.
- The outer jacket is made of either PVC or Teflon.
- Inside the jacket are Kevlar strands to strengthen the cable.
- Kevlar is a strong material used in the fabrication of bulletproof vests.
- Below the Kevlar is another plastic coating to cushion the fiber.
- The fiber is at the center of the cable, and it consists of cladding and core.

Cable Composition



Light Sources

- LED(Light Emitting Diode)
  - Cheap
  - Unfocused light
  - Uncontrollable
  - For short-distance use

- ILD(Injection Laser Diode)
  - Focused light
  - Long distance

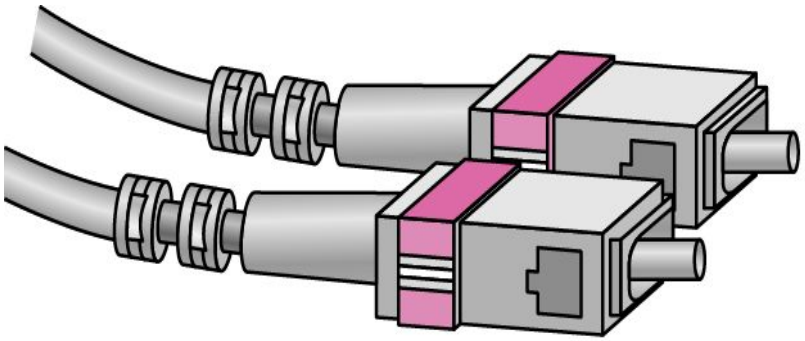
Fiber-optic cable connectors

- The **subscriber channel (SC)** connector is used for cable TV.
- It uses a push/pull locking system.

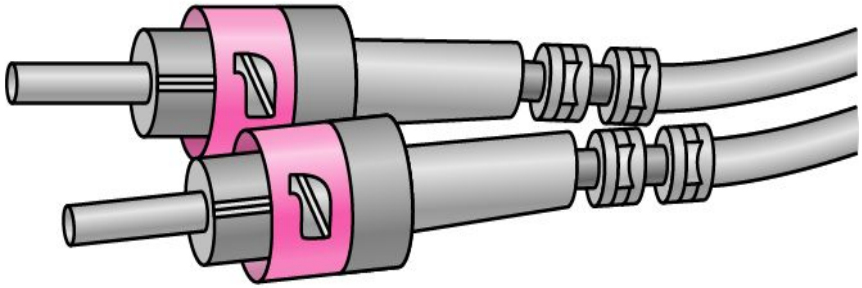
- The **straight-tip (ST)** connector is used for connecting cable to networking devices.
- It uses a bayonet locking system and is more reliable than SC.

- **MT-RJ** is a connector that is the same size as RJ45.

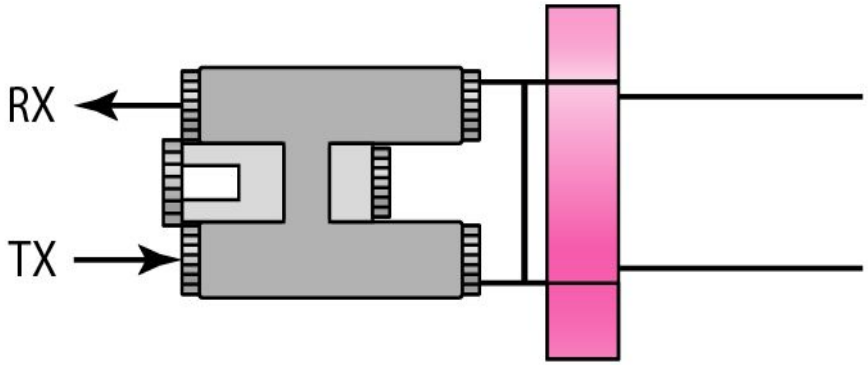
Fiber-optic cable connectors



SC connector



ST connector



MT-RJ connector

Applications of Optical Fiber

- Fiber-optic cable is often found in backbone networks because its wide bandwidth is cost-effective.
- Today, with wavelength-division multiplexing (WDM), we can transfer data at a rate of 1600 Gbps.
- The SONET network provides such a backbone.
- Some cable TV companies use a combination of optical fiber and coaxial cable, thus creating a hybrid network.
- Optical fiber provides the backbone structure while coaxial cable provides the connection to the user premises.
- This is a cost-effective configuration since the narrow bandwidth requirement at the user end does not justify the use of optical fiber.
- Local-area networks such as 100Base-FX network (Fast Ethernet) and 1000Base-X also use fiber-optic cable.

Advantages of Optical Fiber

1. Higher bandwidth
2. Less signal attenuation
3. Immunity to electromagnetic interference
4. Resistance to corrosive material
5. Light weight
6. Greater immunity to tapping

Disadvantages of Optical Fiber

1. Cost
2. Installation/Maintenance
3. Fragility
4. Unidirectional light propagation

Summary

- Transmission Media and its Types
- Guided Media
  - Twisted Pair
    - UTP
    - STP
  - Coaxial Cable
  - Optical Fiber
    - Multimode
    - Single mode

Suggested Reading

- Section
  - 7.1,

    » “Data Communications and Networking”4th Edition by Behrouz A. Forouzan